Methods Supplement: Identity, Classification, and Selection Provenance

From Hairballs to Hypotheses

13 October 2026

This note accompanies the shortened manuscript. It preserves classification and identity detail and incorporates the repository selection audit supplied on 9 October 2026. It has not yet been incorporated into a new Zenodo deposit.

1 Deduplication and Related Reports

Deduplication consolidated 187,446 source occurrences into 140,959 bibliographic records. A *source occurrence* is an entry returned by a provider or imported from another source; several occurrences may refer to the same publication. A *canonical record* is the representative bibliographic entry retained after applying the identity rules.

Matching used normalized DOI keys where available and exact normalized-title keys otherwise. Records with and without DOIs were kept separate when title agreement alone did not establish their identity. arXiv identity adjudications were incorporated into the final total. Source occurrences and identity decisions were retained so that canonical records could be traced to their acquisition routes.

Fifty related-version links connect publications such as preprints and published articles. These links preserve relationships between reports that remain separate bibliographic records. Grouping reports about the same software system occurs during full-text assessment, as described in the main paper.

2 Provisional Candidate Classification

The task groups, assistance modes, and visualization modalities are defined in Section 3 of the main paper. Each label was assigned a Present, Absent, or Uncertain state, and multiple labels could coexist. Classification used evidence for the specific category: for example, unspecified display information yielded uncertainty about modality. The same category definitions were applied to all 9,505 advanced records using their supplied titles and abstracts.

These provisional classifications organized the candidate pool and guided selection for full-text assessment. Table 1 summarizes assistance–modality co-occurrences among candidate records. A record contributes to a cell when both labels are Present; cells overlap. In the export, Present and Uncertain labels are listed explicitly, with omission denoting Absent under the classification contract. Desktop/Planar, for example, was Present in 2,912 records and Uncertain in 4,072.

The table describes labels supported by bibliographic metadata. Establishing that a particular assistance mechanism operates within a particular display environment requires assessment of the system description, which underlies Figure 3 of the main paper.

Table 1: Machine-coded candidate co-occurrences among 9,505 records. Both labels must be PRESENT; cells overlap and are not additive. These are discovery counts, not verified mechanism placements. D: Desktop/Planar; L: Large Display; A: AR/MR.

Assistance	D	L	VR	A	CAVE
Algorithmic	2,837	37	511	399	11
Adaptive	806	12	185	190	2
Conversational	86	1	15	22	0
Immersive	112	20	428	346	8

3 Purpose-Specific Selection Rules

The 384 selected report identities represent six sets and 381 systems, not 384 completed assessments or one uniform sample. Of the report identities, 376 link exactly to the terminal 9,505-record pool. Three additional systems entered through coauthor-supplied reports and five through bounded early map expansion. A distinct prior-survey-only full-report selection route is not established by the cumulative crosswalk.

3.1 Initial 20 entries

Purposive anchors informed by earlier model outputs and discussion were combined with assessed coauthor reports to establish the mechanism map. The candidate export resolved identities of already chosen examples, rather than ranking them. The lead author supplied the twelve anchor nominations; a Codex agent nominated the replacement and seven additions from coauthor-source evidence. The initial 20 was a resulting set, not a documented numerical cap. The intellectual origin of the supplied anchors and any formal ordering or tie-breaking rule remain unresolved.

3.2 Bounded 18 entries

A frozen queue excluded previously represented systems and prioritized reusable lawful local reports, machine-coded candidates for empty cells, sparse Large Display/AR/MR/CAVE cases with explicit mechanism leads, and bounded contrast cases. Per-entry rationales and queue order are retained. The lead author specified the priorities and a maximum of 40; a Codex agent authored the exact 18-entry queue. Within-priority tie rules and the reason for stopping at 18 remain undocumented.

3.3 Expanded 50 entries

Software ranked candidates after excluding prior identities. Scores reflected rare modalities, non-desktop Adaptive/Conversational labels, mechanism/evaluation terms, contrast terms, and access indicators; ordering used descending score then lowercase title. The final 50 were a separately specified list with per-entry rationales, not the ranked top 50. Software validated and materialized that list within a maximum of 60. A Codex agent curated the final nominations under the lead author's maximum of 60. Rules for choosing among similarly suitable records and the reason for selecting exactly 50 remain undocumented.

3.4 Deterministic 200 entries

Software validated the 9,505-record input and excluded previously considered canonical IDs, exact titles and DOIs, and subsequent exact DOI/title duplicates. Scores combined sparse-cell coverage, workflow, life-science, interaction, computation, evaluation/mechanism terms, abstracts, likely open access, assistance/modality bonuses, and contrast indicators. The implemented

ordering was high-precision signal, rationale class, likely-open-access status, descending score, then canonical identifier. The lead author authorized the 200-entry bound and limit of 140 completed new assessments. The first 200 were frozen: 1–140 were primary and 141–200 ordered reserves. Retrieval continued through queue exhaustion; the maximum of 140 concerned completed new assessments rather than selected report identities. The original audit JSON's textual sort description omits the high-precision and open-access strata. A dated correction overlay, `queue_selection_audit_correction_20261009.json` in the same staging directory, records the implemented order and binds it to the executable script. The original audit remains unchanged; no selection was rerun and no queue membership, order, or assessment changed.

3.5 Sparse-cell 24 entries

An upstream 25-entry shortlist from an 8,832-record partial export allowed up to three candidates per sparse cell, favored abstracts and DOIs, and excluded prior exact IDs/titles. Software rebound the entries to the terminal 9,505 export and checked canonical IDs, DOI, normalized title, and related versions. Removing AREA 2013 as a related report of previously considered AREA 2012 yielded 24 entries in preserved priority order, within a maximum of 30. The lead author supplied the provisional 25-entry list and authorized the maximum of 30; software performed reconciliation. The original intellectual source of the supplied list and within-cell tie rules remain unknown.

3.6 Complete target-cell pass: 72 new entries

Software enumerated every exact Present-labelled record in Immersive × Desktop/Planar (112 memberships), Immersive × Large Display (20), and Adaptive × Large Display (12). The 144 memberships represented 131 records, including 58 prior exact identities. All entered processing; priority band, descending metadata score, and canonical identifier determined access order only. Identity reconciliation reduced them to 129 groups. Excluding already represented systems and related reports yielded 72 new systems in the cumulative selected set. The lead author specified and authorized the three target cells. The 72 new entries were the reconciliation outcome, not an inclusion quota.

4 Access and System-Level Reconciliation

Set	Reports	Systems	Full text	Abstract	Unavailable	Assessed
20	20	20	20	0	0	20
18	18	18	15	0	3	15
50	50	50	25	0	25	25
200	200	197	60	1	139	57
24	24	24	1	0	23	1
72	72	72	14	0	58	14
Total	384	381	135	1	248	132

Two related-report pairs and an iCAVE preprint linked to an already represented system account for the report–system difference. At the September cutoff, the 132 completed assessments comprise 80 eligible, 19 unresolved, and 33 excluded/contextual systems. Unavailable reports remain access outcomes rather than scientific exclusions.

An October access follow-up supplied three previously unavailable reports within the same selected set: VIPER (Algorithmic × Desktop/Planar), NAVRNA (Immersive × Large Display, supported by tangible constrained deformation), and Cosmic View (Algorithmic × Large Display, using a planetarium dome). [60, 8, 20] These author-confirmed assessments bring the totals to 138 accessible report entries, 245 unavailable entries, one abstract-only entry, and 135

completed system assessments (83 eligible, 19 unresolved, and 33 excluded/contextual). Three additional placements bring Figure 3 to 79 represented systems and 122 placements. The six original selection sets and their membership were not rerun or expanded. The September table above remains the original dated reconciliation.

5 Trace and Authority Limits

The report–system crosswalk at `evidence/selection-flow/unit_crosswalk.csv` records assessment set, queue identifier, route, canonical linkage, report identity and version, access, system outcome, and placement qualifications. Wave-specific scripts, frozen queues, source bindings, retrieval manifests, and assessment ledgers provide supporting trace; dedicated execution logs were not found for the static queues.

Contemporaneous session evidence identifies the requesting user's direction of priorities, bounds, and target cells and distinguishes Codex nominations from deterministic software operations. Morris Chukhman retrospectively confirmed these roles on 9 October 2026. The repository record at `docs/full_report_selection_authority.md`, linked from `docs/reproducibility.md`, separates this retrospective confirmation from contemporaneous evidence and preserves unresolved shortlist origins.

Nomination authority, automated assessment, later author review, and independent validation are distinct. No separate author approval of the exact constructed queues before assessment is documented. The lead author has reported reviewing and confirming the Figure 3 placements; this later review supplies neither dated row-level approval records nor independent double coding or pool-wide screening validation. The crosswalk does not provide a reason for every other candidate's non-selection.

6 Bounded Language-Evidence Follow-up

The initial system assessment used accessible English-language reports. Language was not a provider-query or title-and-abstract screening gate. A subsequent audit distinguished provider-reported language, title/abstract language signals, and full-report language; an English abstract was not treated as proof of an English report. The 9,505-record candidate export contains no language column.

A 147-report lead set was ranked for bounded author review. Seven active Tier A reports were assessed; Trellet's French thesis remains separately pending passage verification by Amira Kefi. Marchianò's thesis, Chemoface, and Campoalegre's thesis have English bodies despite their bibliographic language signals; Le Baher's thesis, REDPath, AIMA, and Wang's VisConnectome report have French, German, Spanish, and Chinese bodies, respectively. Six reports received provisional eligibility recommendations, including a qualified infrastructure-boundary case; VisConnectome's substantive computational-assistance criterion remains unresolved. These are assistant recommendations pending author adjudication, not seven approved systems or additions to the manuscript's counts. Companion/version relationships and consequential translated passages require verification. This purposive follow-up does not estimate screening reliability or establish exhaustive non-English coverage.

7 Task-Symbol Interpretation

Task letters identify navigation (N), selection/filtering (F), comparison (C), sensemaking (S), and coordination/collaborative reasoning (R). Coding follows descriptions of the implemented analytical workflow, not a demonstrated benefit or an interface affordance alone. Coordination can involve task assignment, handoff, or collective interpretation; it need not imply synchronous

multiuser operation. Mechanism–environment placement evidence and task support are distinct. Retained placement rows include analytical-task descriptions, source locators, and qualifications. Repeated task letters do not establish equivalent support, performance, or evaluation across systems.

8 Per-System Evidence for the Supported Map

Table 2 consolidates the 79 mapped systems and 122 supported assistance–modality pairings in Figure 3. It includes the October access follow-up and Vitessce Link's measurement-based Immersive × AR/MR amendment. Pairings and their task letters remain separate within each system; they do not imply support for other combinations. The four eligible systems without supported placements are outside this mapped-system table.

Entries summarize retained report evidence and assessment notes, not new independent validation. Generic scientist-role descriptions do not establish specific inspect/change/undo controls. Unless stated, report limitations have not been separately extracted; placement qualifications are assessment boundaries rather than necessarily author-reported limitations. Missing or ambiguous evaluation information is distinguished from an explicitly unreported evaluation. Seven reports use retained titles or identifiers because their bibliography keys remain unresolved. Row identifiers refer to the companion per-system CSV's evidence locators.

Table 2: Per-system evidence for the retained supported Figure 3 map. Tasks: N navigation, C comparison, F selection, S sensemaking, R collaboration. Pairings are explicit; missing evidence is labelled, not inferred.

System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S001 5-D stem-cell correction workflow [69]	Report application: Visualization and correction of automated segmentation, tracking and lineaging from 5-D stem cell image sequences.. Tasks: Algorithmic × Desktop/Planar: NCFS	Automated segmentation, tracking, and lineage construction are coupled to visualization and explicit analyst correction across 5-D stem-cell sequences.	Algorithmic × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	Retained abstract claims correction to 100% accuracy; validation design and independent outcomes cannot be established from that claim.	S001 (CSV locators).
S002 AEGIS. Exploratory Gene Ontology Analysis with Interactive Visualization `10.1038/s41598-019-42178-x`	Report application: Exploratory Gene Ontology Analysis with Interactive Visualization. Tasks: Algorithmic × Desktop/Planar: NCFS	A focus-plus-context graph layout, ontology traversal, simulations, and power studies organize GO relations and support biological investigation.	Algorithmic × Desktop/Planar	The scientist selects GO concepts and study questions; the system lays out, simulates, and computes; the scientist interprets and revises the inquiry.	Information not yet extracted: evaluation design and findings.	S002 (CSV locators). Preprint is the assessed accessible version; DOI links the published report.
S003 Aneurysm CBR visual analytics [58]	Report application: Combining visual analytics and case-based reasoning for rupture risk assessment of intracranial aneurysms.. Tasks: Algorithmic × Desktop/Planar: NCFS	Case-based reasoning computes similarity rankings that are linked to morphology and clinical-feature views.	Algorithmic × Desktop/Planar	Clinician selects features/weights and reviews retrieved cases; system ranks similarities; clinician assesses rupture evidence.	Six trained physicians completed evaluation tasks; limited sample and training context constrain outcome claims.	S003 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S004 AR knee-replacement visualization [42]	Report application: A Novel Visualization System of Using Augmented Reality in Knee Replacement Surgery: Enhanced Bidirectional Maximum Correntropy Algorithm.. Tasks: Algorithmic × AR/MR: NCFS	CT segmentation, stereo matching, WLS occlusion handling, and bidirectional maximum-correntropy ICP registration drive the report's AR overlay workflow.	Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Retained passage describes execution-time measurement and a comparison figure; metric definitions, baseline and outcomes unresolved.	S004 (CSV locators). Implemented on sampled CT images and online surgical videos; this does not establish intraoperative clinical benefit.
S005 ARGV [19]	Report application: ARGV: 3D genome structure exploration using augmented reality.. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × AR/MR: NCFS	The web system transforms chromosome-conformation data into interactive 3D genome structures exposed on both planar and AR displays.	Algorithmic × Desktop/Planar; Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	A comparative user study is mentioned in the retained abstract; participants, design and numerical results not yet extracted.	S005 (CSV locators).
S006 ASCRIBE-XR [47]	Report application: Turning Immersive Viewers into Analytical Workspaces: ASCRIBE-XR and Agent-Driven Scientific Visualization.. Tasks: Algorithmic × VR: NFSR; Adaptive × VR: NFSR; Conversational × VR: NFSR	The implemented agent invokes segmentation, meshing, resolution, and rendering tools inside the immersive workflow. — The scientific request and loaded data are triggers; the agent resolves unstated method/resolution choices and changes its tool sequence. — Open-ended language specifies a scientific intent that the agent operationalizes across multiple analysis and rendering tools.	Algorithmic × VR; Adaptive × VR; Conversational × VR	Scientist states a goal; the agent resolves operational choices and executes a workflow; scientist inspects and can refine the output.	Implemented architectural demonstrations, including a Panicum hallii root use case; no participant study or scientific-outcome comparison.	S006 (CSV locators). Implemented examples exceed parameter editing, although benefit was not evaluated. — The mechanism resolves analytic operations rather than mapping speech to a fixed command.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S007 Augmented Reality Microscope [30]	Report application: Independent assessment of a deep learning system for lymph node metastasis detection on the Augmented Reality Microscope. Tasks: Algorithmic × AR/MR: CFS	A deep-learning metastasis detector supplies case-level overlays through an augmented-reality microscope while pathologists inspect lymph-node tissue.	Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Pathologist re-review of selected false-negative fields and annotation errors; no analytical-benefit conclusion inferred.	S007 (CSV locators).
S008 BioNet-XR [54]	Report application: BioNet-XR: Biological Network Visualization Framework for Virtual Reality and Mixed Reality Environments. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × VR: NCFS	Desktop BioNet-XR computes layouts, shortest paths, and PageRank/Steiner/ random-walk subnetworks. — The VR version implements analysis functions similar to the desktop build for network exploration.	Algorithmic × Desktop/Planar; Algorithmic × VR	Analyst chooses nodes/algorithms; software computes layouts, paths, and subnetworks; analyst interprets network structure.	Desktop-only usability study with 30 participants; VR/MR benefit was not tested because participants saw videos only.	S008 (CSV locators).
S009 BioNetApp [50]	Report application: BioNetApp: An interactive visual data analysis platform for molecular expressions.. Tasks: Algorithmic × Desktop/Planar: NCFS	Correlation, comparison, clustering, outlier detection, and curve fitting are computed within coordinated molecular-expression views.	Algorithmic × Desktop/Planar	Analyst selects experiments and filters; algorithms mine correla- tions/clusters; analyst inspects time-course patterns.	Alcohol-induced fatty-liver case study; no participant or baseline study.	S009 (CSV locators).
S010 CellWhisperer [52]	Report application: Multimodal learning enables chat-based exploration of single-cell data.. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	A multimodal language model maps dialogue about single-cell data to analytical operations and visual results.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	Retained passage introduces biological-characteristic prediction tests; design and outcomes not yet extracted. Chat availability is not proof of analytic benefit.	S010 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S011 CERA [5]	Report application: ChatGPT-Enhanced ROC Analysis (CERA): A shiny web tool for finding optimal cutoff points in biomarker analysis. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	The Shiny application computes ROC cutoffs and uses an LLM to mediate explanations and analysis choices.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S011 (CSV locators).
S012 CoCoNUT [3]	Report application: CoCoNUT: an efficient system for the comparison and analysis of genomes.. Tasks: Algorithmic × Desktop/Planar: CFS	Rare matches, chaining, alignment, and comparative plots reveal genomic similarity, rearrangement, and duplication.	Algorithmic × Desktop/Planar	Analyst sets the comparative problem and parameters; algorithms generate chains/alignments; analyst inspects genomic structure.	Benchmarks and comparisons with BioMart and prior synteny results; no user study.	S012 (CSV locators).
S013 Compressed Adjacency Matrices [17]	E. coli and B. subtilis regulatory networks; Inspect motifs, paths, and neighborhoods in gene-regulatory networks. Tasks: Algorithmic × Desktop/Planar: NCFS	Structure-aware cutting, ordering, and compression reorganize a gene-regulatory-network matrix before interactive highlighting, filtering, and path or neighborhood inspection.	Algorithmic × Desktop/Planar	Inspect motifs, paths, and neighborhoods in gene-regulatory networks. Interaction consequence: Reorganizes and navigates the representation	Network demonstrations and informal biologist feedback Limitations: No controlled insight study; report doubts routine use by biologists; benefit boundary: Qualitative structural visibility	S013 (CSV locators). The algorithm structures the view; interaction does not revise the organizing algorithm.
S014 Cosmic View [20]	Hierarchical biological relationships and similarities. Tasks: Algorithmic × Large Display: NCS	Constrained multidimensional scaling and spherical decluttering organize biological similarities for inspection.	Algorithmic × Large Display	Scientist explores organized biological relationships. Detailed scoping, editing and undo controls not yet extracted.	Information not yet extracted: evaluation design, participants, findings and report limitations are not in the retained follow-up row or mechanism paragraphs.	S014 (CSV locators). Large Display pairing is the planetarium dome; no extra Desktop, CAVE or Immersive pairing inferred.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S015 CPA. CPA: a web-based platform for consensus pathway analysis and interactive visualization `10.1093/nar/gkab421`	Report application: CPA: a web-based platform for consensus pathway analysis and interactive visualization. Tasks: Algorithmic × Desktop/Planar: NCFS	Eight pathway methods, cross-dataset meta-analysis, and consensus computation feed interactive pathway and gene-relation views.	Algorithmic × Desktop/Planar	The analyst chooses methods and datasets; CPA computes and combines results; the analyst compares pathways and interprets biological meaning.	Information not yet extracted: evaluation design and findings.	S015 (CSV locators). Established methods are integrated substantively; no Adaptive mechanism is inferred.
S016 D-Rax [1]	Report application: D-Rax: Domain-specific Radiologic assistant leveraging multi-modal data and eXpert model predictions. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	The radiologic assistant combines images with expert-model predictions and mediates diagnostic exploration through language.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	Retained passage includes a truncated numerical table and discussion of expert-model-guided training; exact metrics/comparisons cannot be reliably recovered here.	S016 (CSV locators).
S017 DatAR [64]	Report application: Exploring Neuroscience Literature and Understanding Relations Between Brain-Related Topics - Using Augmented Reality. Tasks: Algorithmic × AR/MR: NCFS	Literature-derived brain-topic relations are organized as a graph and explored through an AR brain representation.	Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Retained passage describes neuroscientist interviews and iterative relation-finding design; exact sample and outcome measures not yet extracted.	S017 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S018 DTBIA [74]	Report application: DTBIA: An Immersive Visual Analytics System for Brain-Inspired Research. Tasks: Algorithmic × VR: NCFS; Immersive × VR: NCFS	3D force-directed edge bundling, top-connection selection, threshold filtering, animation, and hierarchical region-to-voxel-to-slice analysis organize functional and structural brain data. — The system couples real-scale and large-scale brain models with walking, flying, teleportation, and controller-based spatial navigation while computational filtering and bundling guide multiscale inspection.	Algorithmic × VR; Immersive × VR	Information not yet extracted: scientist-role controls.	Expert cases are mentioned in retained manuscript prose; sample, design and findings not yet extracted. No controlled estimate of an immersive advantage.	S018 (CSV locators). The paper evaluates two expert case studies, not a controlled comparison of scientific outcomes. — Immersive placement rests on multiscale spatial navigation coupled to filtering and bundling, not on the headset alone.
S019 Echo [66]	Report application: Echo: A Large Display Interactive Visualization of ICU Data for Effective Care HandOffs. Tasks: Algorithmic × Large Display: NCFSR	Patient summaries, normalcy scores, and threshold-based automatic anomaly detection are presented on a tiled wall for team comparison, annotation, and handoff discussion.	Algorithmic × Large Display	Clinicians share inspection of computed patient summaries and anomaly highlighting during ICU handoff; detailed computational scoping/edit/undo controls not yet extracted.	Information not yet extracted: evaluation design and findings.	S019 (CSV locators). Algorithmic assistance is simple threshold comparison, not Adaptation; evaluation is small and does not establish clinical outcome benefit.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S020 ExaViz [18]	Report application: ExaViz: a flexible framework to analyse, steer and interact with molecular dynamics simulations. Tasks: Algorithmic × Desktop/Planar: NCFS; Adaptive × Desktop/Planar: NCFS; Algorithmic × CAVE: NCFSR; Adaptive × CAVE: NCFSR; Immersive × CAVE: NCFSR	Desktop ExaViz combines molecular-dynamics analysis, content-aware camera paths, semi-automatic viewpoints, and interactive steering. — User-selected molecular targets and geometry drive least-occluded viewpoints and task-aware paths; recorded steering targets can drive a later automatic run. — The same molecular computations, guided navigation, analysis graphs, and steering operate in the implemented CAVE-like environment. — Target selection and molecular geometry tailor viewpoints and paths; a captured human steering path controls later automatic steering with manual correction. — Two head-tracked views, co-located pointing, spatial atom picking, and a haptic device that follows the user couple assistance to embodied CAVE state.	Algorithmic × Desktop/Planar; Adaptive × Desktop/Planar; Algorithmic × CAVE; Adaptive × CAVE; Immersive × CAVE	The scientist selects targets and initiates steering; the system proposes views and executes/replays forces; the scientist observes, overrides, corrects, and interprets.	Information not yet extracted: evaluation design and findings.	S020 (CSV locators). The framework demonstrates technical capability; controlled scientific benefit is not established. — Data/task-conditioned tailoring and interaction-history replay qualify; ordinary parameters alone are not the basis. — One system occupies multiple modalities; this is not a second system count. — Implemented feedback/history loop; “adapt” wording about human behavior is not itself the evidence. — Immersive placement rests on tracked spatial assistance and haptics, not CAVE display alone.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S021 Facetto [34]	Multiplex tissue images and cell features; Discover and correct cell phenotypes. Tasks: Algorithmic × Desktop/Planar: NCFS; Adaptive × Desktop/Planar: NCFS	EM clustering, UMAP, and CNN classification are integrated with coordinated image, projection, distribution, table, and phenotype-tree views. — Experts select and label phenotype subsets; clustering results train a classifier whose later assignments change, creating a user-feedback-to-model loop.	Algorithmic × Desktop/Planar; Adaptive × Desktop/Planar	Discover and correct cell phenotypes. Interaction consequence: Changes clusters/labels and feeds revised labels into later classification	Ten interviews and two guided expert case studies Limitations: Small samples; authors operated UI; qualitative outcomes; tree lacks fuzzy membership; benefit boundary: Experts could direct phenotype analysis in cases	S021 (CSV locators). Published identity is linked to an accessible related preprint; active learning is future work. — Adaptive placement rests on demonstrated classifier retraining, not on parameter adjustment; experts directed cases while authors operated the interface.
S022 FathomGPT [32]	FathomNet observations and images; Explore ocean-science data through natural-language intentions. Tasks: Algorithmic × Desktop/Planar: FS; Conversational × Desktop/Planar: FS	The system resolves scientific names, constructs and executes SQL or Python, retrieves results, and generates interactive charts from user requests. — Natural-language subgoals and conversation history mediate query construction, result retrieval, and chart refinement in the planar interface.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Explore ocean-science data through natural-language intentions. Interaction consequence: Changes queries, displays, and returned data	2,977 workshop prompts; 781-prompt success analysis; context ablation Limitations: Not a controlled scientific-decision or adoption study; phrasing and data limitations; benefit boundary: 72.08% prompt success; context improved correctness in reported ablation	S022 (CSV locators). Prompt-level success is not evidence of improved scientific decisions. — Conversation supports bounded subgoals; it does not establish durable goal learning or autonomous scientific agency.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S023 Flud [9]	Biological interaction networks; Produce biologically useful network layouts. Tasks: Algorithmic × Desktop/Planar: FSR	A multi-criterion layout score, algorithmic clues, and simulated annealing alternate with crowd node placements to improve biological-network layouts.	Algorithmic × Desktop/Planar	Produce biologically useful network layouts. Interaction consequence: Human moves and algorithm search alter the layout; no preference learning	Roughly 2,000 crowd workers and seven biologically trained participants in published study Limitations: Layout/task metrics do not demonstrate biological discovery; benefit boundary: Higher layout scores and task effectiveness	S023 (CSV locators). The system does not learn an individual user's preferences; layout improvements do not establish better biological discovery.
S024 focusedMDS / distnet [68]	Report application: Focused multidimensional scaling: interactive visualization for exploration of high-dimensional data.. Tasks: Algorithmic × Desktop/Planar: CFS	FocusedMDS and distnet recompute local projections and distance relations around analyst-selected objects.	Algorithmic × Desktop/Planar	Analyst focuses objects and compares views; algorithms recompute local projections/networks; analyst judges distortion and structure.	Worked biological examples; no participant study.	S024 (CSV locators).
S025 Fused DTI/HARDI. Fused DTI/HARDI Visualization	Report application: Fused DTI/HARDI Visualization. Tasks: Algorithmic × Desktop/Planar: NCFS	A classification scheme automatically selects DTI or HARDI per voxel; GPU glyphs and tractography expose the fused result interactively.	Algorithmic × Desktop/Planar	The analyst chooses views and encodings; the system selects voxel models and renders; the analyst interprets anatomy and can adjust display controls.	Qualitative user evaluation retained; design/sample/outcomes not yet extracted. Scientific-outcome improvement not established.	S025 (CSV locators). Qualitative user evaluation supports utility, not scientific-outcome improvement.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S026 Gene Tracer [39]	Report application: Gene Tracer: a smart, interactive, voice-controlled Alexa skill For gene information retrieval and browsing, mutation annotation and network visualization.. Tasks: Algorithmic × Desktop/Planar: NCFS	Gene retrieval, mutation annotation, and network construction are integrated with an interactive planar visualization; the voice layer remains a boundary case.	Algorithmic × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S026 (CSV locators).
S027 Generic anatomical-model builder [73]	Report application: Building generic anatomical models using virtual model cutting and iterative registration. Tasks: Algorithmic × Desktop/Planar: CFS	Interactive model cutting defines subvolumes that rigid, affine, and deformable registration algorithms align and average.	Algorithmic × Desktop/Planar	Analyst specifies the subvolume; software executes registration and averaging; analyst interprets group-shape differences.	No participant study; registration sharpness, Dice similarity, and RMSE are reported.	S027 (CSV locators).
S028 GENET. Development and Validation of an AI-Driven System for Automatic Literature Analysis and Molecular Regulatory Network Construction	Report application: Development and Validation of an AI-Driven System for Automatic Literature Analysis and Molecular Regulatory Network Construction. Tasks: Algorithmic × Desktop/Planar: NCFS	NLP named-entity and relation models construct a filterable molecular-regulatory knowledge graph for mechanism search and hypothesis generation.	Algorithmic × Desktop/Planar	The system extracts relations and constructs the graph; the researcher searches, filters, interprets mechanisms, and validates selected hypotheses experimentally.	Information not yet extracted: evaluation design and findings.	S028 (CSV locators). Model accuracy and one checked pathway support capability, not broad decision benefit.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S029 GeneVAnD [29]	Report application: Visualization methods for statistical analysis of microarray clusters.. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × Large Display: NCFSR	Statistical cluster summaries and linked gene views are computed and coordinated on conventional screens. — The same linked analytical views are shown and used together on a display wall.	Algorithmic × Desktop/Planar; Algorithmic × Large Display	Analyst chooses clusters and linked views; statistical routines compute summaries; analyst interprets gene patterns.	Worked microarray examples; no controlled user study.	S029 (CSV locators).
S030 i-gRINN [55]	Report application: i-gRINN: a next-generation web server for protein energy network analysis of heterogeneous biomolecular systems with natural language-based data exploration.. Tasks: Algorithmic × Desktop/Planar: NCFS; Conversational × Desktop/Planar: NCFS	Protein-energy networks are computed and queried through a web server whose natural-language layer mediates data exploration.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	Acknowledgements mention beta testing and feedback; no formal evaluation design or outcomes extracted.	S030 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S031 iCAVE [38, 37]	Report application: iCAVE: an open source tool for visualizing biomolecular networks in 3D, stereoscopic 3D and immersive 3D — iCAVE: an open source tool for immersive 3D visualization of complex biomolecular interaction networks. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × CAVE: NCFS	The same 3D layout, clustering, edge-bundling, network-query, and metric functions are analytically presented in a desktop screen environment. — 3D layout, clustering, edge bundling, network metrics, and database-generated networks are presented and explored in a projection-based CAVE environment.	Algorithmic × Desktop/Planar; Algorithmic × CAVE	Information not yet extracted: scientist-role controls.	Information not yet extracted: evaluation design and findings.	S031 (CSV locators). Desktop/Planar is the frozen modality for rotatable 3D shown on a planar screen; stereoscopic desktop glasses do not create a sixth modality. — Established as Algorithmic × CAVE. Immersive assistance is not inferred because the report does not show the assistance algorithm changing with tracked CAVE context.
S032 iGEAK [10]	Report application: iGEAK: an interactive gene expression analysis kit for seamless workflow using the R/shiny platform.. Tasks: Algorithmic × Desktop/Planar: CFS	An interactive Shiny pipeline computes normalization, differential expression, clustering, and enrichment views.	Algorithmic × Desktop/Planar	Analyst explicitly chooses methods and parameters; the pipeline recomputes results; analyst compares and interprets them.	Worked datasets and workflow demonstration; no controlled user study.	S032 (CSV locators).
S033 Interactive 3D lesion mapping [14]	Report application: Neurosurgical Craniotomy Localization Using Interactive 3D Lesion Mapping for Image-Guided Neurosurgery. Tasks: Algorithmic × AR/MR: NCFS	Image registration and 3D lesion mapping transform patient scans into an interactive AR localization view.	Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S033 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S034 Interactive AI annotation in VR [46]	Report application: Interactive AI annotation of medical images in a virtual reality environment.. Tasks: Algorithmic × VR: NCFS; Adaptive × VR: NCFS; Immersive × VR: NCFS	A deep model converts sparse corrective clicks into a revised whole-volume segmentation in VR. — Radiologist corrective clicks are the trigger; the learned segmentation changes across the volume after each feedback cycle. — 3D click locations in the immersive volume become positive/negative model prompts that alter segmentation.	Algorithmic × VR; Adaptive × VR; Immersive × VR	Radiologist supplies corrective examples; the model revises a volume-wide segmentation; radiologist reviews and corrects the result.	Three radiologists, 50 scans, 192 annotation tasks; quality improved for two radiologists but not the third; unlocalized changes remained a concern.	S034 (CSV locators). This feedback-to-model loop exceeds direct local editing. — The spatial input changes computation, not just camera pose.
S035 LIM Tracker [7]	Report application: LIM Tracker: a software package for cell tracking and analysis with advanced interactivity. Tasks: Algorithmic × Desktop/Planar: NCFS	Automatic tracking results are visualized, corrected, and rerun in a linked microscopy workflow.	Algorithmic × Desktop/Planar	Analyst inspects and corrects tracks; the system reruns tracking; analyst validates lineage continuity.	Worked cases and SEG/DET/TRA benchmark metrics; no controlled analyst study.	S035 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S036 Liver MR navigation [72]	Report application: Comprehensive mixed reality surgical navigation system for liver surgery.. Tasks: Algorithmic × AR/MR: NCFS; Adaptive × AR/MR: NCFS; Immersive × AR/MR: NCFS	Rigid registration and LIBR deformation correction transform intraoperative surface evidence into an updated liver model. — New tracked intraoperative surface data are the trigger; the registration intervention changes the preoperative model displayed in HoloLens. — Tracked stylus/headset coordinates and the local surgical surface resolve spatial correspondence and update the overlaid model.	Algorithmic × AR/MR; Adaptive × AR/MR; Immersive × AR/MR	Surgeon supplies a surface or tracked points; registration updates the model; surgeon interprets the corrected spatial relation.	Phantom validation across five deformation states; no clinical outcome study.	S036 (CSV locators). This is data/context-conditioned model correction, not direct parameter adjustment. — Spatial context determines the analytic registration/correction, beyond maintaining a view.
S037 MediSyn [28]	Linked CGI and DTC biomedical datasets; Compare drugs, targets, tumor types, provenance, and conflicting evidence. Tasks: Algorithmic × Desktop/Planar: CFS	Linked biomedical datasets are algorithmically synthesized into matrix layers; sorting, uncertainty encodings, and provenance links support comparison and evidence inspection.	Algorithmic × Desktop/Planar	Compare drugs, targets, tumor types, provenance, and conflicting evidence. Interaction consequence: Changes evidence presentation and ordering; not model state	Six-person counterbalanced drug-selection and inconsistency tasks Limitations: Small sample, simplified tasks, limited domain familiarity; no clinical-decision endpoint; benefit boundary: Faster task completion with correct answers in study	S037 (CSV locators). The six-person study supports simplified evidence-selection tasks, not improved treatment decisions.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S038 Metis [43]	Generated molecules and medicinal-chemistry feedback; Teach molecular generators detailed expert preferences. Tasks: Algorithmic × Desktop/Planar: FS; Adaptive × Desktop/Planar: FS	The GUI shows generated molecules, similar active compounds, and model explanations while converting structured feedback into reward-model or reward-function inputs. — Chemist feedback trains a reward model or updates a constructed reward function; a REINVENT loop reloads newly generated molecules for further feedback.	Algorithmic × Desktop/Planar; Adaptive × Desktop/Planar	Teach molecular generators detailed expert preferences. Interaction consequence: Changes feedback data, reward model, and later generated outputs	Technical implementation and fictitious MAPK10 examples Limitations: No participant or outcome experiment; desktop and REINVENT coupling; benefit boundary: No evaluated benefit reported	S038 (CSV locators). The report demonstrates the mechanism with fictitious examples and no participant or outcome study. — Adaptive placement rests on feedback changing model scoring and later outputs, not on simple parameter editing; alignment benefit is proposed rather than evaluated.
S039 MinOmics [41]	Report application: MinOmics, an Integrative and Immersive Tool for Multi-Omics Analysis.. Tasks: Algorithmic × Large Display: NCFS; Algorithmic × VR: NCFS	Database queries, network analysis, and structural comparison are jointly displayed on an 8.3 m^2 wall. — The redox-proteomics use case superposes selected protein structures in a VR headset for fold comparison.	Algorithmic × Large Display; Algorithmic × VR	Analyst selects datasets and structural subsets; queries and models are computed; analyst compares folds and formulates biological interpretations.	Performance benchmarks and a scientific use case; formal user studies explicitly not yet conducted.	S039 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S040 Molecular Dynamics Visualization (MDV) [70]	Report application: Molecular Dynamics Visualization (MDV): Stereoscopic 3D Display of Biomolecular Structure and Interactions Using the Unity Game Engine.. Tasks: Algorithmic × Desktop/Planar: NCS; Algorithmic × Large Display: NCS	MDV loads trajectories and recomputes secondary-structure representations in real time on its desktop build. — The same trajectory and secondary-structure computations drive the HIVE cylindrical stereoscopic display.	Algorithmic × Desktop/Planar; Algorithmic × Large Display	Analyst loads and manipulates molecular trajectories; Stride recomputes secondary structure; analyst interprets changing conformations.	Technical performance and application cases; no comparative analyst-outcome study.	S040 (CSV locators).
S041 moleculARweb Virtual Modeling Kits [4, 13]	Report application: Building blocks for commodity augmented reality-based molecular visualization and modeling in web browsers. — Visualization, Interactive Handling and Simulation of Molecules in Commodity Augmented Reality in Web Browsers Using moleculARweb's Virtual Modeling Kits.. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × AR/MR: NCFS	Implemented browser tools compute molecular mechanics and experimental observables while users manipulate or navigate molecular models in the documented planar or AR workflow.	Algorithmic × Desktop/Planar; Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S041 (CSV locators). Implemented prototype research tasks are supported; broader research utility and benefit remain unevaluated.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S042 MR bone-tumor planning [71]	Report application: Mixed Reality Improves 3D Visualization and Spatial Awareness of Bone Tumors for Surgical Planning in Orthopaedic Oncology: A Proof of Concept Study. Tasks: Algorithmic × AR/MR: NCFS	Patient-specific bone-tumor models are reconstructed and manipulated in MR to compare spatial relationships and surgical plans.	Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S042 (CSV locators).
S043 MR carotid-endarterectomy planning [12]	Report application: Mixed reality applied to surgical planning and tailoring of carotid endarterectomies.. Tasks: Algorithmic × AR/MR: NCFS	Patient-specific vascular models are segmented and manipulated in MR to compare anatomy and tailor the operative plan.	Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S043 (CSV locators).
S044 MR radiation-dose visualization [63]	Report application: Mixed Reality Visualization of Radiation Dose for Health Professionals and Patients in Interventional Radiology.. Tasks: Algorithmic × AR/MR: CFS	Radiation-dose calculations are registered to the interventional scene so clinicians can inspect and compare exposure patterns in MR.	Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S044 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S045 MR thalamic-surgery workflow [75]	Report application: Role of mixed reality in the workflow of complex neurooncological surgeries: a case analysis in thalamic surgery.. Tasks: Algorithmic × AR/MR: NCFS	Multimodal imaging and segmented tumor/anatomy models are combined in MR for thalamic-surgery planning and workflow review.	Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Retained abstract describes a retrospective 2022–2024 surgical cohort with clinical/radiological data; sample and outcomes not yet extracted.	S045 (CSV locators).
S046 NAVRNA [8]	Report application: From Visualization to Manipulation of RNA Secondary and Tertiary Structures. Tasks: Immersive × Large Display: FSR	Tracked tabletop tokens specify moved nodes and anchors; constrained deformation preserves edge lengths and handles collisions.	Immersive × Large Display	Scientist-role details not extracted	Evaluation details not extracted	S046 (CSV locators). Immersive pairing limited to embodied constraints on deformation, not pan/zoom or display size. October target_group_id differs from persisted September group; exact canonical ID/DOI binds this report.
S047 NeuroBlocks [6]	Large electron-microscopy volumes and segmentation states; Coordinate segmentation and proofreading across large connectomics projects. Tasks: Algorithmic × Desktop/Planar: NFR	Automatic RhoANA segmentation is integrated with manual and semi-automatic proofreading, task assignment, comparison, provenance, approval, and restoration views.	Algorithmic × Desktop/Planar	Coordinate segmentation and proofreading across large connectomics projects. Interaction consequence: Changes segments, task assignments, approvals, and restored workflow states	Two neuroscientist collaborations including a 955-GB dataset and proofreading case Limitations: Qualitative evidence; no controlled productivity or scientific-outcome comparison; benefit boundary: Qualitative utility in cases	S047 (CSV locators). Qualitative case studies do not quantify productivity or scientific-outcome gains.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S048 Next Viewer [27]	Report application: Next viewer–development of a software for the spatial computing data glasses apple vision pro for patient-specific preoperative planning and navigation. Tasks: Algorithmic × AR/MR: NCFS	Patient-specific imaging is segmented and spatially rendered in Apple Vision Pro for preoperative inspection and planning.	Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Retained evaluation passage reports overall score +1.7 and participant feedback. Scale, design and sample not yet extracted; latency, learning curve and workflow integration limits are mentioned.	S048 (CSV locators).
S049 NUI-VR2 [26]	Vessel and tumor volumes; Delineate and inspect medical volumes. Tasks: Algorithmic × VR: NCF; Immersive × VR: NF	User-placed 3D seeds feed SVM-based feature selection and segmentation that creates a probability volume for assisted volume rendering. — The segmentation pipeline consumes spatial seeds placed with in-environment gestures and returns a probability volume that users inspect in the same immersive 3D workspace.	Algorithmic × VR; Immersive × VR	Delineate and inspect medical volumes. Interaction consequence: Changes segmentation parameters and rendered outputs; not adaptive learning	Mouse-versus-NUI initialization on vessel and four tumors using Dice/Hausdorff Limitations: No participant study or separate usability/decision-benefit measure; benefit boundary: Technical segmentation comparison	S049 (CSV locators). Speech commands are an input channel, not evidence of Conversational or Adaptive assistance. — Immersive placement rests on spatial seed specification coupled to segmentation, not on VR display alone; no participant study establishes usability or clinical decision benefit.
S050 OncoProExp [49]	Report application: OncoProExp: An interactive shiny web application for comprehensive cancer proteomics and phosphoproteomics analysis. Tasks: Algorithmic × Desktop/Planar: CFS	Statistical and pathway analyses are integrated with interactive cancer proteomics views.	Algorithmic × Desktop/Planar	Analyst selects cohorts and analyses; software computes differential/pathway results; analyst interprets molecular patterns.	Worked cancer datasets and technical validation; no controlled analyst-benefit study.	S050 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S051 PAE Viewer [21]	Report application: PAE viewer: a webserver for the interactive visualization of the predicted aligned error for multimer structure predictions and crosslinks.. Tasks: Algorithmic × Desktop/Planar: NCFS	Linked predicted-error, sequence, structure, and cross-link views support inspection of protein-model reliability.	Algorithmic × Desktop/Planar	Analyst selects residues/regions and cross-links; system coordinates uncertainty and structure views; analyst judges model reliability.	Worked protein examples; no controlled user study.	S051 (CSV locators).
S052 Pan-cancer networks. Efficient exploration of pan-cancer networks by generalized covariance selection and interactive web content `10.1093/nar/gkv413`	Report application: Efficient exploration of pan-cancer networks by generalized covariance selection and interactive web content. Tasks: Algorithmic × Desktop/Planar: NCFS	Generalized sparse inverse covariance selection integrates cancers and data types, then exposes inferred networks as linked interactive web content.	Algorithmic × Desktop/Planar	The analyst selects cancers and network regions; the system infers and exposes links; the analyst interprets modules against pathway and drug evidence.	Information not yet extracted: evaluation design and findings.	S052 (CSV locators). Method robustness and rediscovered mechanisms do not establish user-benefit causality.
S053 PathwayMatrix [15]	Report application: PathwayMatrix: visualizing binary relationships between proteins in biological pathways.. Tasks: Algorithmic × Desktop/Planar: NCFS	Ordering, grouping, hierarchy, and lenses reorganize pathway relationships for comparative inspection.	Algorithmic × Desktop/Planar	Analyst selects pathways and lenses; software orders and groups relations; analyst compares patterns and traces candidates.	Feedback from three domain experts; a new cross-pathway relation was found, while binary decomposition limits causal/temporal reasoning.	S053 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S054 Patient-health dashboard [53]	Report application: A modular visual analytics dashboard for patient health data. Tasks: Algorithmic × Desktop/Planar: NCFS; Adaptive × Desktop/Planar: NCFS	Transformer LLMs extract and classify clinical entities, which feed interactive medication, laboratory, and highlighted-text modules. — The dashboard dynamically selects modules from document content type and clinical relevance, and the clinician can correct extraction errors.	Algorithmic × Desktop/Planar; Adaptive × Desktop/Planar	The clinician initiates record review and corrects extraction; the system extracts, categorizes, and selects modules; the clinician interprets the resulting history.	Synthetic-record demonstrations; no structured clinical evaluation retained. Clinical benefit not established.	S054 (CSV locators). Implemented capability; synthetic records and no structured clinical evaluation limit benefit claims. — Data-conditioned tailoring qualifies; future user-context personalization is not claimed.
S055 PeCaX [22]	Somatic variants, clinical annotations, pathways, drugs, and literature; Annotate variants and develop treatment hypotheses. Tasks: Algorithmic × Desktop/Planar: NCFSR	ClinVAP variant annotation, pathway and drug-network extraction, and automatic layout feed interactive tables and networks used in molecular-tumor-board inspection.	Algorithmic × Desktop/Planar	Annotate variants and develop treatment hypotheses. Interaction consequence: Changes evidence selection, annotation, and presentation	Processing-time benchmark Limitations: No quantitative usability or clinical-decision study; usability study is future work; benefit boundary: Processing performance only	S055 (CSV locators). Processing performance is reported; a usability study is future work.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S056 PhenoFlow [33]	Clinical cohort exploration. Tasks: Algorithmic × Desktop/Planar: NCFS; Conversational × Desktop/Planar: NCFS	An LLM synthesizes executable cohort-construction code from metadata; temporal folding and linked cohort and patient views expose results for supervision and exploration. — Neurologists define and refine cohorts in natural language; the LLM normalizes terminology, synthesizes executable operations, and returns explanations and small-multiple diagnostics for supervision.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Neurologist specifies a natural-language cohort request and can inspect generated wrangling code, explanations and field views. Scientific interpretation remains with the analyst; undo controls not yet extracted.	Design collaboration with two neurologists and clinical case studies during iterative refinement; no controlled estimate of calibration improvement. Explanations could conceal errors or omit needed information.	S056 (CSV locators). Not placed as Adaptive: changing generated code in response to a direct request is not by itself system-selected adaptation. — Natural language mediates analytic operations and explanation; the authors describe the LLM as fragile and keep neurologists in the supervisory loop.
S057 Protein-model validation. A Visual Environment for Data Driven Protein Modeling and Validation	Report application: A Visual Environment for Data Driven Protein Modeling and Validation. Tasks: Algorithmic × Desktop/Planar: NCFS	Validation metrics and local quality measures are computed and coordinated across heatmap, molecular, and statistical views.	Algorithmic × Desktop/Planar	The expert chooses the model and validation questions; metrics guide attention; the expert inspects linked evidence and decides what to repair.	Retained placement qualification explicitly reports no controlled user-benefit study; examples demonstrate capability only.	S057 (CSV locators). Examples demonstrate capability; no controlled user-benefit study is reported.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S058 ProteinSketch [40]	Protein backbones, volumes, sequences, structures, and binders; Express three-dimensional topology and envelope constraints for protein design. Tasks: Algorithmic × VR: NFS; Immersive × VR: NFS	RFdiffusion, ProteinMPNN, structure prediction, and filtering refine user-specified protein topologies and volumetric envelopes. — Bare-hand bimanual sketching encodes position, orientation, curvature, connectivity, and volumetric envelopes as explicit spatial constraints for generative models.	Algorithmic × VR; Immersive × VR	Express three-dimensional topology and envelope constraints for protein design. Interaction consequence: Changes generative constraints, model inputs, and generated/refined outputs	Computational and experimental tests across topologies, geometry, expression, and binding Limitations: Preprint; no usability/adoption study; fidelity/designability tradeoffs; benefit boundary: Design feasibility and experimental expression	S058 (CSV locators). Preprint status must remain explicit; biological feasibility is not a user-performance evaluation. — Immersive placement rests on VR-native spatial constraint specification; no learned user-preference mechanism is shown.
S059 Rapid QC-MS [51]	Report application: Rapid QC-MS: Interactive Dashboard for Synchronous Mass Spectrometry Data Acquisition Quality Control. Tasks: Algorithmic × Desktop/Planar: CFS	Each new file is automatically processed, classified against QC criteria, and shown in a synchronous interactive dashboard.	Algorithmic × Desktop/Planar	Researcher defines QC criteria; software processes each new file and classifies status; researcher diagnoses drift and decides whether to intervene.	Operational demonstrations and instrument benchmarking; no controlled user study.	S059 (CSV locators).
S060 RATS VR planning system [67]	Report application: Developing a Virtual Reality Simulation System for Preoperative Planning of Robotic-Assisted Thoracic Surgery.. Tasks: Algorithmic × VR: NCFS	Patient CT reconstructions are segmented and rendered as manipulable VR anatomy for relation checking and resection planning.	Algorithmic × VR	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S060 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S061 REDAC [16]	Report application: REDAC: RNA-seq expression data analysis chatbot.. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	A chatbot invokes RNA-seq expression analysis and returns plots and statistical results for iterative inspection.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S061 (CSV locators).
S062 RetainVis [35]	Large claims datasets for heart failure and cataract; Inspect and steer longitudinal clinical-risk predictions. Tasks: Algorithmic × Desktop/Planar: CFS	An interpretable recurrent model produces risk estimates and feature contributions that linked views expose for cohort and patient-history exploration and user-directed what-if steering.	Algorithmic × Desktop/Planar	Inspect and steer longitudinal clinical-risk predictions. Interaction consequence: Changes input records, contributions/model state, and predictions	Predictive comparisons, expert feedback, and fictitious case study Limitations: Authors state user problem-solving benefit was not empirically tested; benefit boundary: Predictive performance comparisons	S062 (CSV locators). Not placed as Adaptive: reported changes follow explicit edits and retraining commands, without evidenced system-selected tailoring; user benefit was not empirically tested.
S063 RNA-SeqEZPZ [65]	Bulk RNA-seq datasets; Run RNA-seq analysis from FASTQ through differential expression and pathways. Tasks: Algorithmic × Desktop/Planar: CFS	A point-and-click interface launches a fixed RNA-seq workflow for quality control, alignment, counting, differential expression, and pathway analysis, then exposes interactive result views.	Algorithmic × Desktop/Planar	Run RNA-seq analysis from FASTQ through differential expression and pathways. Interaction consequence: Changes run parameters and presentation; not learned model state	Public/new datasets and output comparisons Limitations: No controlled user, decision-quality, adoption, or maintenance study; benefit boundary: Technical operation and output comparison	S063 (CSV locators). Commanded pipeline execution, not Adaptive assistance; evaluation does not establish improved scientific decisions.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S064 Robin [11]	Report application: Robin: an advanced tool for comparative loop caller result analysis leveraging large language models.. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	Loop-caller outputs are computed, compared, and explained through an LLM-mediated web workflow.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S064 (CSV locators).
S065 SAMIRA [57]	Report application: Towards user-centered interactive medical image segmentation in VR with an assistive AI agent. Tasks: Algorithmic × VR: NFS; Adaptive × VR: NFS; Conversational × VR: NFS; Immersive × VR: NFS	Foundation-model segmentation, retrieval-augmented guidance, and 3D rendering execute operations requested by the user. — Guidance is conditioned on the currently viewed slice, and user point prompts revise masks and propagate corrections. — Natural-language dialogue is interpreted into localization, explanation, segmentation, refinement, and visualization operations. — Head, gaze, and controller pointing specify corrections in a true-scale VR workspace while the agent updates the segmentation and guidance.	Algorithmic × VR; Adaptive × VR; Conversational × VR; Immersive × VR	The user asks, points, and corrects; the agent localizes, segments, explains, and renders; the user approves and interprets the final 3D result.	Usability evaluation and controller/head/eye input comparison retained; clinical effectiveness not established. Participant details and outcomes not yet extracted.	S065 (CSV locators). Implemented; usability is evaluated, not clinical effectiveness. — Adaptation rests on context and corrective feedback, not simple parameter entry. — Dialogue mediates analytic subgoals rather than button-equivalent commands. — Spatial input is computationally coupled to segmentation and guidance; HMD display alone is not the evidence.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S066 scSelector [25]	Report application: scSelector: A Flexible Single-Cell Data Analysis Assistant for Biomedical Researchers.. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	Natural-language requests invoke single-cell filtering, comparison, and visualization operations whose results remain available for researcher inspection.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	PBMC case analysis compares seven rare cells with a control cluster using differential expression and enrichment; no controlled user benefit inferred.	S066 (CSV locators).
S067 SeqPlots [62]	Report application: SeqPlots - Interactive software for exploratory data analyses, pattern discovery and visualization in genomics.. Tasks: Algorithmic × Desktop/Planar: CFS	Server-side aggregation generates heatmaps and profiles that remain linked to interactive genomic filters.	Algorithmic × Desktop/Planar	Analyst chooses regions, groups, and filters; system aggregates aligned genomic signals; analyst compares patterns.	Software validation and worked use cases; no participant study.	S067 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S068 Skin-lesion AR [24]	Report application: An Augmented Reality Mobile Application for Skin Lesion Data Visualization. Tasks: Algorithmic × AR/MR: CS; Immersive × AR/MR: CS	Image preprocessing, lesion segmentation, feature extraction, photometric stereo, and CNN classification produce parameters and a melanoma classification shown in a mobile AR view. — Spatially grounded assistance detects device-to-skin distance, centers and selects a lesion, adjusts illumination, computes a classification, and overlays the result in the dermatologist's view.	Algorithmic × AR/MR; Immersive × AR/MR	Information not yet extracted: scientist-role controls.	Retained placement qualification reports CNN accuracy 74.6% and a preliminary evaluation with seven dermatologists. No clinical-effectiveness result established; design and further outcomes not yet extracted.	S068 (CSV locators). No repository canonical ID was found; CNN accuracy was 74.6% and the preliminary evaluation involved seven dermatologists. — Immersive placement rests on tracked spatial context and in-situ overlay, not on phone display alone; no clinical-effectiveness result is established. — Canonical report ID not recorded in the historical ledger; identity is retained through persisted group, exact title and bibliography/DOI. No canonical ID invented.
S069 Space-Time Hypercube [23]	Report application: Immersive and interactive visualization of 3D spatio-temporal data using a space time hypercube: Application to cell division and morphogenesis analysis.. Tasks: Algorithmic × VR: NCFS; Immersive × VR: NCFS	The STH projection, clipping, filtering, lineage propagation, and coordinated views transform embryo time-series data in VR. — A controller-resolved spatial cell selection drives propagation through time and lineage and synchronizes both VR representations.	Algorithmic × VR; Immersive × VR	Analyst positions cuts and selects cells; the system propagates state through lineage/time and linked views; analyst detects temporal events.	Five biologists used two one-hour sessions; wave detection was useful, but other tasks and workflows were mixed.	S069 (CSV locators). The counted mechanism is spatial selection feeding a computed propagation, not VR display alone.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S070 SpinalSAM-R1 [2]	Report application: SpinalSAM-R1: A Vision-Language Multimodal Interactive System for Spine CT Segmentation. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	Vision-language prompts condition spine CT segmentation and return inspectable masks in an interactive planar workflow.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Generic assessment; specific analyst controls not extracted.	Retained passage reports evaluation on 120 lumbar CT scans (31,454 slices); baseline metrics and human-analytic outcomes not yet extracted.	S070 (CSV locators).
S071 Surgical Theater XR platform [61, 56]	Report application: Extended reality platform for minimally invasive endoscopic evacuation of deep-seated intracerebral hemorrhage: illustrative case. — Novel planning pipeline utilizing the Surgical Theater system for pediatric epilepsy surgery.. Tasks: Algorithmic × VR: NCFS; Algorithmic × AR/MR: NCFS	Segmented patient anatomy and tractography are assembled into interactive 360-degree models for VR planning and AR intraoperative reference.	Algorithmic × VR; Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	One retained clinical case describes patient-specific XR-supported planning; no comparative causal benefit inferred or transferred to all companion reports.	S071 (CSV locators).
S072 Toxygates [45]	Report application: Interactive Toxicogenomics: Gene set discovery, clustering and analysis in Toxygates.. Tasks: Algorithmic × Desktop/Planar: CFS	Selected gene sets are clustered, displayed as heatmaps, and passed to enrichment routines for mechanistic toxicogenomics analysis.	Algorithmic × Desktop/Planar	Analyst defines gene sets and cutoffs; system clusters and computes enrichment; analyst builds a mechanistic interpretation.	Mechanistic case study; no controlled user evaluation; dataset-normalization limits are acknowledged.	S072 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S073 VIADS [31]	Report application: A visual interactive analytic tool for filtering and summarizing large health data sets coded with hierarchical terminologies (VIADS).. Tasks: Algorithmic × Desktop/Planar: NCFS	VIADS aggregates hierarchical terms and computes filtered summaries/comparisons with interactive preview and threshold revision.	Algorithmic × Desktop/Planar	Analyst selects algorithm and thresholds; software aggregates and compares hierarchies; analyst inspects summaries and revises thresholds.	Implementation report and algorithm examples; usability and utility studies were future work.	S073 (CSV locators).
S074 VIPER [60]	Patient-specific MRI/atlas integration for functional neurosurgery. Tasks: Algorithmic × Desktop/Planar: NCFS	Automated atlas registration aligns anatomical structures to patient MRI; linked planar/3D views support inspection.	Algorithmic × Desktop/Planar	Scientist inspects registered anatomy, proposed lesions and surrounding structures. Registration editing/undo controls not yet extracted.	Information not yet extracted: evaluation design, participants, findings and report limitations are not in the retained follow-up row or mechanism paragraphs.	S074 (CSV locators). Desktop/Planar includes its stereoscopic monitor; stereo display alone is not Immersive assistance.
S075 Virtual embolization [59]	Report application: Virtual embolization for treatment support of intracranial AVMs using an interactive desktop and VR application.. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × VR: NCFS	Precomputed embolization and flow simulations are selected and compared in a desktop application. — The VR application exposes the same simulated occlusion alternatives for spatial comparison.	Algorithmic × Desktop/Planar; Algorithmic × VR	Clinician chooses embolization scenarios; simulation produces candidate flow outcomes; clinician compares consequences.	Two neuroradiologists provided formative feedback; they valued virtual blocking but requested quantitative information.	S075 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S076 Virtual Island. Bringing the Field into the Lab: Large-Scale Visualization of Animal Movement Trajectories within a Virtual Island `10.1109/LDAV48142.2019.8944350`	Report application: Bringing the Field into the Lab: Large-Scale Visualization of Animal Movement Trajectories within a Virtual Island. Tasks: Algorithmic × CAVE: NCFS	Parallel computation filters individuals and time spans while integrated terrain, canopy, and GPS trajectories are explored in CAVE2.	Algorithmic × CAVE	The researcher chooses animals and time spans; parallel filtering executes; the researcher navigates the CAVE2 scene and interprets behavior in environmental context.	Information not yet extracted: evaluation design and findings.	S076 (CSV locators). Short conference report demonstrates capability, not comparative analytical benefit.
S077 Vitessce Link [44]	Report application: Vitessce Link: A Mixed Reality and 2D Display Hybrid Approach for Visual Analysis of 3D Tissue Maps. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × AR/MR: NCFSR; Immersive × AR/MR: CFS	Linked selections and multiscale tissue-map computations coordinate a planar Vitessce view with registered MR representations. — Tracked finger positions specify measurement endpoints; geometric distance is computed within the tissue map.	Algorithmic × Desktop/Planar; Algorithmic × AR/MR; Immersive × AR/MR	Scientist specifies hand-position endpoints for tissue measurement and inspects linked planar/MR views. Detailed undo and intervention controls not yet extracted.	Kidney and melanoma case studies and user/community feedback are retained; these do not independently establish improved analytical performance.	S077 (CSV locators). — Measurement-based Immersive pairing only; synchronization, navigation and collaboration alone are not the basis. No Adaptive or Conversational addition or independently validated performance gain.
S078 VROOM [36]	Report application: Virtual reality for the observation of oncology models (VROOM): immersive analytics for oncology patient cohorts.. Tasks: Algorithmic × VR: NCFS	Patient-cohort variables and oncology models are computationally arranged for immersive VR comparison and filtering.	Algorithmic × VR	Generic assessment; specific analyst controls not extracted.	A case-study walkthrough is mentioned in the retained conclusion; evaluation design and findings not yet extracted.	S078 (CSV locators).

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S079 VR–AR medical-image analysis workflow [48]	Report application: Enhanced Medical Image Analysis: Leveraging 3D Visualization and VR-AR Technology. Tasks: Algorithmic × VR: NCFS; Algorithmic × AR/MR: NCFS	Implemented preprocessing, volume/surface reconstruction, rendering, slicing, and gesture interaction transform medical images for VR/AR inspection.	Algorithmic × VR; Algorithmic × AR/MR	Generic assessment; specific analyst controls not extracted.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	S079 (CSV locators). Capability is reported in a preprint; participant and model-evaluation reporting is limited and not independently replicated.

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